A Static-Universe Two-Channel Photon Propagation Model Confronted with DES-SN5YR

A Standalone Adversarial Test of Redshift, Temporal Remapping, Chromatic Transfer, Surface Brightness, and Energy Closure

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Abstract

This paper asks a deliberately adversarial question: can the principal Type Ia supernova observables normally interpreted as evidence for metric expansion be reproduced by a non-expanding Euclidean spatial geometry with two photon–vacuum propagation channels, while surviving the classical tests that eliminate simpler tired-light models? "Static" here means no large-scale metric expansion; Version 7 explicitly allows source populations and cosmic contents to evolve with emission epoch and therefore does not assume a stationary eternal galaxy ensemble.

The first propagation channel is coherent and direction preserving. At effective level it has two observable faces that must ultimately be derived from one microscopic vertex. An inelastic frequency-shift face gives $d\ln\nu/dD = -K/c$, hence $D = (c/K)\ln(1+z)$ for steady and transient sources. A distinct reactive group-delay face responds to transient illumination above the saturated ambient corridor. In the unsaturated window its group-index perturbation is written $n_g = 1 + \kappa_{\rm r}K\xi$, with $\xi = t - r/c$ measured behind the transient front. The leading front remains on $n_g = 1$ while a trailing independent photon obeys $d\ln\Delta t/dD = \kappa_{\rm r}K/c$, giving $\Delta t_{\rm obs} = \Delta t_{\rm em}(1+z)^{\kappa_{\rm r}}$. DES-SN5YR measures $b = 1.003 \pm 0.005\,({\rm stat}) \pm 0.010\,({\rm sys})$, requiring $\kappa_{\rm r} \simeq 1$. The kinematics are therefore explicit and causal without optical coherence between photons; the unresolved microscopic problem is to derive the equality between the reactive rate and the inelastic drag rate from the vacuum action rather than impose it phenomenologically.

A second, non-forward channel has $\sigma_{\rm nf} \propto E^{1/2} \propto \lambda^{-1/2}$. The physically defined rest-B asymptotic optical depth corresponding to the DES fit is $\tau_{\infty,B} = 0.594$, removing 10.3% of the coherent beam at $z = 0.5$ and 16.0% at $z = 1$. On the released DES-SN5YR Hubble diagram, the resulting one-parameter static distance law is nearly degenerate with one-parameter flat ΛCDM: $\chi^2 \simeq 1641.1$ versus 1640.1 on the default 1,768-object diagnostic sample, $\Delta\chi^2 = +1.1$, with both $\chi^2/{\rm dof} \simeq 0.93$.

Several tempting supporting mechanisms were then tested rather than retained. A full Hsiao-SED × measured-DECam-filter calculation shows that Wien/passband slicing plus the chromatic opacity changes fitted stretch by at most 5×10^{-4} and peak epochs by at most 0.05 d when the temporal remap is disabled; it cannot manufacture the observed $b \simeq 1$. A preregistered iron-group secondary-maximum clock measured on raw DES i/z photometry gives $b_2 = 0.52 \pm 0.78$ from 120 detections: explicitly inconclusive, consistent with both universal remapping and no dilation because DES cadence and the $z \lesssim 0.3$ visibility window are underpowered.

The strongest present empirical threat is the Tolman surface-brightness test. The model predicts $D_A = D = (c/K)\ln(1+z)$ and $I \propto (1+z)^{-2}e^{-\tau_\lambda}$. Re-expressing the published Lubin–Sandage distance sensitivity in the static frame removes the nominal I-band crisis but moves tension to R; the approximately frame-invariant observed spread $n_I - n_R = 0.78 \pm 0.21$ differs from the model prediction $\simeq -0.02$ by about 3.7σ before a full static-frame re-reduction. The paper also closes the previously omitted forward-energy ledger: at $z = 1$ the coherent frequency shift transfers 50% of each photon's emitted energy out of the observed photon, while the non-forward branch removes a further 16% of the surviving beam in rest B. The proposed equilibrium-bath interpretation of the CMB therefore owes both FIRAS-level thermalization and the observed CMB anisotropy/BAO correlation structure. Version 7 does not claim those debts are solved. Its claim is narrower and falsifiable: the static supernova alternative is sufficiently specified to confront data, has survived some canonical objections, has failed some proposed mechanisms, and now has a finite set of experiments capable of killing it.

1 Thesis, scope, and terminology

The thesis tested in this paper is:

> **Thesis.** A non-expanding Euclidean spatial geometry with two photon–vacuum propagation channels can reproduce the observed achromatic redshift, the approximately $(1+z)$ temporal broadening of supernova signals, and the DES-SN5YR luminosity-distance relation without a dark-energy acceleration term. The same transfer law must then survive independent tests of chromaticity, angular distance, surface brightness, radiance, image sharpness, source-population evolution, and energy conservation.

Three distinctions are load bearing.

First, *static geometry is not stationary contents.* Light still has finite travel time. For the distance law derived below,

$$t_{\rm lb}(z) = \frac{D(z)}{c} = K^{-1}\ln(1+z). \tag{1}$$

Version 7 therefore adopts the logically available branch "static spatial geometry, evolving contents." Observed stellar-population evolution, the cosmic star-formation history, quasar evolution, and related redshift trends are not switched off; they become constraints on the history of matter and sources evaluated against Eq. (1). This costs any claim that the observed universe is a strictly stationary eternal ensemble.

Second, the model is not classical tired light. Classical energy-loss models generally predict no macroscopic time dilation and often introduce chromaticity or image blur. Here redshift, transient timing, and non-forward chromatic depletion are separate observables with separate failure modes.

Third, the paper follows an anti-confirmation-bias rule. A mechanism that fails a registered test is retained as a null result, not reinterpreted. An underpowered measurement is labeled inconclusive. An adverse observable is placed in the same status ledger as favorable ones. The public repository preserves preregistration amendments, implementation defects, and superseded mechanisms.

2 Physical motivation without microscopic overclaiming

Nonlinear electromagnetic propagation is not a forbidden category. The Euler–Heisenberg effective action predicts nonlinear QED interactions, and SLAC E-144 observed nonlinear Compton processes and multiphoton pair production using an intense laser and a high-energy electron beam (Burke et al., 1997; Bamber et al., 1999). Those experiments do not establish the cosmological interaction proposed here, and they do not supply its rate. They establish only that a nonlinear vacuum electromagnetic response is physically meaningful.

The present object is therefore an *effective propagation model*. Its coefficients are constrained by astronomical observations. A successful microscopic theory must derive them from the underlying vacuum dynamics while respecting spectral-line coherence, image sharpness, polarization, local laboratory constraints, energy conservation, and the CMB. Where that derivation is absent, the paper says so explicitly.

3 Effective two-face forward vertex

3.1 Inelastic coherent face: redshift for every photon

The coherent forward interaction is assumed direction preserving and achromatic in fractional energy loss. For every photon, steady or transient,

$$\frac{dE}{dD} = -\frac{K}{c}E, \qquad \frac{d\nu}{dD} = -\frac{K}{c}\nu. \tag{2}$$

Integrating gives

$$1 + z = \frac{\nu_{\text{em}}}{\nu_{\text{obs}}} = e^{KD/c}, \qquad \boxed{D(z) = \frac{c}{K}\ln(1+z)}. \tag{3}$$

The same fractional shift applies to all spectral coordinates,

$$\lambda_{\text{obs}} = (1+z)\lambda_{\text{em}}. \tag{4}$$

A thermal spectral shape therefore has its Wien peak mapped to the coordinate temperature $T/(1+z)$. This statement concerns spectral coordinates only; full Planck radiance normalization is a separate test in Section 9.

Equation (2) is the redshift face of the effective vertex. It does not require a preconditioned corridor and remains active for steady galaxy light. It also creates a substantial energy ledger: by redshift z, the photon has transferred the fraction

$$f_{\text{fwd}}(z) = 1 - \frac{1}{1+z} = \frac{z}{1+z} \tag{5}$$

out of the observed photon. At $z = 1$, $f_{\text{fwd}} = 0.5$. Section 11 treats the destination of this energy as a required closure condition.

3.2 Reactive face: a transient shared group-delay wake

The temporal problem is different. A time-translation-invariant, unchanged medium cannot stretch the arrival interval between mutually incoherent photons emitted days apart. The model therefore assigns a second, reactive face to the same effective interaction: a weak group-delay perturbation produced by a *transient change in illumination above the saturated ambient corridor*.

Let

$$\xi = t - \frac{r}{c} \tag{6}$$

measure observer time behind the advancing transient front. In the linear unsaturated response window write the group index

$$n_g(\xi) = 1 + \kappa_{\text{r}} K\xi, \qquad 0 < \xi < T_{\text{sat}}, \tag{7}$$

with a saturated ambient state outside the transient ramp. The coefficient $\kappa_{\rm r}$ measures the reactive rate relative to the inelastic drag rate K.

The leading transient front sits at $r = ct$, hence $\xi = 0$ and $n_g = 1$ for its entire trajectory. Explicitly,

$$\frac{dn_g}{dt} = \partial_t n_g + \dot{r}\,\partial_r n_g = \kappa_{\rm r} K + c\left(-\frac{\kappa_{\rm r} K}{c}\right) = 0. \tag{8}$$

This is the point that resolves the earlier distance-law objection: the local transient index excursion can be tiny even though the propagation effect accumulates over a cosmological path. The leading front is never slowed to $c/(1+z)$; it rides the moving $\xi = 0$ surface.

Now consider a trailing photon emitted independently a time Δt later. It samples $\xi = \Delta t$ and therefore, to first order,

$$\frac{d\Delta t}{dD} = \frac{\kappa_{\rm r} K}{c}\Delta t. \tag{9}$$

Thus

$$\boxed{\Delta t_{\rm obs} = \Delta t_{\rm em} e^{\kappa_{\rm r} K D/c} = \Delta t_{\rm em}(1+z)^{\kappa_{\rm r}}.} \tag{10}$$

The measured supernova time-dilation exponent is therefore

$$b = \kappa_{\rm r}. \tag{11}$$

White et al. measure (White et al., 2024)

$$b = 1.003 \pm 0.005\ (\text{stat}) \pm 0.010\ (\text{sys}), \tag{12}$$

so the effective response requires $\kappa_{\rm r} \simeq 1$.

This formulation uses no mutual optical coherence between photons. The corridor field is the shared memory variable. It is also more honest than the earlier claim that $b = 1$ was already a microscopic identity: *at effective level* the measured coefficient fixes the equality of the reactive and inelastic rates. Deriving $\kappa_{\rm r} = 1$ from a single microscopic vacuum vertex is still owed.

3.3 What defines $\xi = 0$ in a permanently illuminated universe?

A sharpened referee objection asks what "the transient front" means when the line of sight has been illuminated by the host galaxy and diffuse backgrounds for gigayears. The effective answer is a two-level response:

(i) The inelastic coherent face, Eq. (2), is per photon and always active. It therefore redshifts steady galaxy light without requiring index growth.

(ii) The reactive face responds to the *change* $\delta I(t)$ above a saturated ambient optical state. Ancient steady illumination contributes to the background state but not to an indefinitely growing ramp. For a supernova, $\xi = 0$ is the onset of the supernova's transient increment above that ambient state.

A minimal effective kinetics, not yet a Lagrangian derivation, is of the form

$$\partial_t \chi = 2\kappa_{\rm r} K\, S[\delta I]\, S[\chi_{\rm sat} - \chi] - \frac{\chi}{T_{\rm relax}}, \qquad n_g = 1 + \frac{\chi}{2}, \tag{13}$$

where S is a saturating switch, $T_{\rm relax} \gg$ the supernova duration, and $\chi_{\rm sat} \ll 1$. Equation (13) is a specification of what the microphysics must produce, not evidence that it does. A successful underlying theory must derive the universal intensity-independent drag rate K, the reactive coefficient $\kappa_{\rm r} \simeq 1$, the saturation scale, and the relaxation time from one vacuum response.

This sourcing split also removes a false dichotomy between "global $n(t)$" and "per-wavepacket wakes." The model is neither: redshift is a per-photon inelastic process, while timing is a corridor-level transient response shared by later photons from the same event. Whether one microscopic vertex can actually generate both faces with the required coefficient equality is the central theoretical debt of Version 7.

4 Non-forward chromatic channel

The second propagation channel removes photons from the coherent image-forming beam with an energy-dependent probability. The working scaling is

$$\sigma_{\rm nf}(E) \propto E^{1/2} \propto \lambda^{-1/2}. \tag{14}$$

Along the redshifting path this gives the rest-frame optical depth

$$\tau_\lambda(z) = \tau_{\infty,\lambda}\left[1-(1+z)^{-1/2}\right], \qquad \tau_{\infty,\lambda} = \tau_{\infty,B}\left(\frac{440\,\mathrm{nm}}{\lambda}\right)^{1/2}. \tag{15}$$

The physical magnitude attenuation is

$$\Delta m_\lambda = \frac{2.5}{\ln 10}\tau_\lambda. \tag{16}$$

Earlier versions used a $5/\ln 10$ bookkeeping coefficient; the fitted amplitude is unchanged, but its physical interpretation doubles. The rest-B asymptotic optical depth is therefore

$$\boxed{\tau_{\infty,B} = 0.594}. \tag{17}$$

It removes 10.3% of rest-B coherent-beam photons at $z = 0.5$ and 16.0% at $z = 1$.

The complete observer-frame transfer operator used in the synthetic tests is

$$F_{\rm obs}(\lambda_o, t_o) = A(z)e^{-\tau(\lambda_o/(1+z),z)}S\left(\frac{\lambda_o}{1+z}, \frac{t_o}{(1+z)^{\kappa_{\rm r}}}\right), \tag{18}$$

where $S(\lambda, t)$ is the intrinsic SN spectral-time surface. No factor in Eq. (18) knows which atomic species produced a feature; species physics determines S, while propagation acts on the entire surface. That universality is falsifiable.

5 A mechanism tested and killed: passbands do not fake $b = 1$

A tempting explanation was that Wien displacement, finite observer filters, and phase-dependent chromatic attenuation might make an intrinsically undilated transient appear broader. This was tested directly rather than retained.

Using the Hsiao SN-Ia spectral time series (Hsiao et al., 2007), measured DECam g, r, i, z throughput curves, Eq. (15), and the spectral coordinate shift, we generated synthetic light curves with the temporal remap in Eq. (18) disabled. The recovered stretch changes by at most

$$|\Delta s| \le 5 \times 10^{-4} \tag{19}$$

for $z \le 0.8$ in every band, and recovered peak epochs move by at most 0.05 d. An independently preregistered inverse-injection test likewise returned $\hat{b} = 0$ for time-independent opacity, even when that opacity was amplified far beyond the fitted value. The structural reason is simple: a time-independent multiplicative spectral weight can reshape color and amplitude but does not remap the temporal coordinate.

With the temporal remap enabled at $\kappa_{\rm r} = 1$, the same transfer calculation recovers $(1+z)$ stretch to within 0.05% in each DES band. The chromatic branch therefore survives with a smaller, cleaner job: amplitude and color transfer, not time dilation.

For phase-matched rest-frame windows near 330 and 650 nm the model predicts

$$Q(z) \equiv \frac{[F_{330}/F_{650}]_z}{[F_{330}/F_{650}]_{z\simeq 0}} = e^{-\Delta\tau(z)}, \tag{20}$$

with

$$Q(0.25) = 0.979, \qquad Q(0.5) = 0.964, \qquad Q(1) = 0.944. \tag{21}$$

The propagation component is predicted to be phase-flat at fixed rest wavelengths. Intrinsic line formation and recombination evolve strongly with phase; phase-flatness is therefore part of the discriminant, not merely the amplitude trend.

6 DES-SN5YR distance comparison

6.1 Standalone likelihood definition

The analysis uses the public DES-SN5YR v1.2 Hubble-diagram products, the official $z_{\rm HD}/z_{\rm HEL}$ convention, and the full released STAT+SYS covariance. The absolute-magnitude/distance-scale offset is analytically profiled. The default diagnostic sample excludes entries with effectively zero inverse-variance weight (MUERR > 10), leaving $N = 1768$; retaining all 1,829 released entries changes no quoted headline number.

For the static law,

$$\mu_{\rm stat}(z_{\rm HD}, z_{\rm HEL}) = 5\log_{10}[(1+z_{\rm HEL})\ln(1+z_{\rm HD})] \tag{22}$$

$$+ \frac{2.5}{\ln 10}\tau_{\infty,B}\left[1-(1+z_{\rm HD})^{-1/2}\right] + \mathcal{M}. \tag{23}$$

Flat ΛCDMis fitted with one shape parameter Ω_m and the same profiled offset. The physical opacity prior is non-negative; the numerical fit is bounded well above the optimum.

The released-vector best fits are approximately $\Omega_m = 0.352$ for flat ΛCDMand $\eta = 0.297$ in the historical magnitude-amplitude convention for the static law, equivalent to the physical $\tau_{\infty,B} = 0.594$ used here. The resulting fit values are

$$\chi^2_{\Lambda\rm CDM} \simeq 1640.1, \qquad \chi^2_{\rm stat} \simeq 1641.1, \tag{24}$$

so

$$\boxed{\Delta\chi^2 \equiv \chi^2_{\rm stat} - \chi^2_{\Lambda\rm CDM} = +1.1}, \tag{25}$$

with both $\chi^2/\text{dof} \simeq 0.93$. The maximum offset-removed separation of the two best-fit curves over the DES redshift range is approximately 0.055 mag.

This is a deliberately narrow result. It is a one-dimensional SN distance comparison, not a global cosmological fit. BAO, CMB anisotropies, lensing, growth, nucleosynthesis, and source-population evolution are separate constraints and are explicitly retained in the open ledger below.

6.2 Bias-correction sensitivity is methodological, not cosmological evidence

For completeness, the companion methods analysis reconstructs a diagnostic vector before the event-level DES BBC magnitude correction. The pairwise statistic changes from $+1.1$ on the released vector to $+12.8$ on that diagnostic vector, a total processing leverage of -11.7 χ^2 units. The pre-BBC vector contains known survey-selection bias and the released covariance belongs to the corrected product, so this number is not evidence for the static model. It motivates a matched forward/BBC simulation under the alternative law. The full sign audit, covariance projections, robustness probes, and frozen-result CI live in the independent methods archive DOI 10.5281/zenodo.22022089; no result in the present paper requires accepting a cosmological interpretation of that leverage.

The strongest existing DES counterargument is its explicit reference-cosmology validation. Camilleri et al. reran the bias-correction machinery under multiple reference cosmologies spanning distance-modulus variations of order 0.15 mag and found recovered cosmological shifts subdominant to the DES statistical uncertainty (Camilleri et al., 2024). The offset-removed separation between the released best-fit static curve and best-fit flat ΛCDMis only about 0.055 mag, below that tested amplitude scale. This is relevant evidence that the BBC layer is not arbitrarily fragile. It is not, however, the same experiment as propagating the present non-FLRW transfer law through the simulation and selection pipeline: the published reference set is FLRW-family and the formal validation is shape dependent, not merely an amplitude bound. Version 7 therefore treats the matched alternative-reference rerun as a completion test, not as an assumed failure of DES calibration.

7 Temporal universality and the executed T4 test

White et al. find Eq. (12) from 1,504 DES Type Ia supernovae (White et al., 2024). Any static model with $b = 0$ is therefore excluded. In the present effective model, Eq. (10) makes this a direct measurement of κ_{r} rather than an assumed optical-coherence effect.

Blondin et al. measured multi-epoch spectra of high-redshift SNe Ia and found spectral aging consistent with a $1/(1+z)$ rate (Blondin et al., 2008). This is a physically different clock and rejects no-dilation models, but it still does not distinguish FLRW from a universal non-metric remap.

To add a species-sensitive internal clock, we preregistered a test based on the Type Ia i/z secondary maximum, which is associated with iron-group recombination and opacity/flux redistribution rather than one isolated atomic transition (Jack et al., 2015). The frozen primary regression was

$$y = a + b_2 x + \gamma(x_1 - \bar{x}_1) + \epsilon, \quad y = \ln \frac{\Delta t_{\mathrm{obs}}}{\Delta t_{\mathrm{model}}(z, \mathrm{band})}, \quad x = \ln(1+z). \tag{26}$$

The universal-remap prediction is $b_2 = 1$ and the no-dilation null is $b_2 = 0$.

The executed result is

$$\boxed{b_2 = 0.52 \pm 0.78, \qquad N = 120.} \tag{27}$$

It lies 0.6σ from 1 and 0.7σ from 0. Neither preregistered kill condition fires. The result is therefore *inconclusive*, not evidence for a mixture and not positive evidence for the model.

The run nevertheless established why DES *griz* cannot decide this test. Median per-SN secondary-maximum timing uncertainty was 5.3 d rather than the forecast 2 d; intrinsic bump diversity contributed about an 8 d equivalent spread; and the feature leaves DES *griz* beyond roughly $z = 0.3$. Approximately 25 times the effective statistics would be required for a 3σ 0-versus-1 distinction under the realized noise. The detection-efficiency difference between low- and high-z bins was only 0.8σ.

The complete audit trail is public. Pre-data amendments added an intercept, a frozen GP peak finder, censoring, and a hypothesis-neutral search window. The first executable pipeline then exposed single-epoch difference-imaging artifacts; a post-data 5σ leave-one-out screen and three implementation repairs were disclosed before any valid b_2 result existed. Version 7 keeps that history because the ability to report an unhelpful result is part of the experimental discipline.

8 Chromatic color test already visible in DES

The opacity law predicts an intrinsic redward trend approximately

$$\frac{dc_{\rm prop}}{dz} \simeq +0.019 \text{ mag}/z, \tag{28}$$

with representative $E(B-V) \simeq 0.007, 0.0125, 0.020$ at $z = 0.25, 0.5, 1$ under a simple monochromatic mapping. On 1,635 DES-subset SNe the raw SALT color slope is instead

$$\frac{dc_{\rm raw}}{dz} = -0.0963 \pm 0.0140 \text{ mag}/z. \tag{29}$$

The opposite sign is not presently a clean propagation falsification because the flux-limited DES sample becomes strongly bluer with redshift under its own selection pipeline; the predicted signal is roughly five times smaller than the raw selection trend. The immediate inexpensive comparison is data minus the published FLRW+selection simulation as a function of redshift. A full alternative-reference BBC simulation remains the stronger end-to-end test. Version 7 therefore reports Eq. (29) rather than describing color as an unexamined future observable.

9 Tolman surface brightness and radiance

With static Euclidean geometry, a strictly forward coherent channel, and no transverse magnification from the tiny transient group-index perturbation,

$$D_A = D = \frac{c}{K} \ln(1+z). \tag{30}$$

The energy-redshift and arrival-rate factors give, before non-forward attenuation,

$$D_{L,0} = (1+z)D, \qquad \frac{D_{L,0}}{D_A} = 1+z, \tag{31}$$

not the metric Etherington ratio $(1+z)^2$. Including Eq. (15),

$$I_{\rm obs} \propto (1+z)^{-2} e^{-\tau_\lambda(z)}. \tag{32}$$

Define the effective exponent by $I \propto (1+z)^{-n_{\rm eff}}$,

$$n_{\rm eff}(z,\lambda) = 2 + \frac{\tau_\lambda(z)}{\ln(1+z)}. \tag{33}$$

Near the Lubin–Sandage cluster mean $z \simeq 0.86$, the model predicts approximately $n_R \simeq 2.18$ and $n_I \simeq 2.16$.

Lubin and Sandage analyzed 34 early-type galaxies in three clusters near $z = 0.8$–0.9 and reported $n_R = 2.59 \pm 0.17$ and $n_I = 3.37 \pm 0.13$ in their $q_0 = 1/2$ distance frame (Lubin & Sandage, 2001a,b). They also published the inferred exponents for other q_0 choices, directly exposing the distance-frame sensitivity of the placement on the local surface-brightness–linear-radius relation.

Before retrieving a full per-galaxy static reduction, we preregistered an analytic frame-shift diagnostic using that published q_0 grid. The measured response of the inferred exponent to $\log D_A$ is approximately -3.41/dex in R and -3.17/dex in I. Extrapolating to Eq. (30) moves the nominal values to

$$n_R \simeq 1.45 \pm 0.17, \qquad n_I \simeq 2.30 \pm 0.13. \tag{34}$$

Thus the often-quoted I-band crisis is highly frame dependent, but the problem does not disappear: it migrates to R. More importantly, the two bands move similarly under distance reassignment, leaving the approximately frame-invariant spread

$$(n_I - n_R)_{\mathrm{L\&S}} = 0.78 \pm 0.21 \tag{35}$$

against the model prediction of approximately -0.02. This is a 3.7σ discrepancy before a modern systematic reanalysis.

The analytic extrapolation extends about 3.5 times beyond the D_A range over which the published q_0 grid calibrates the response, so Eq. (34) is not a resolution. The full static-frame reduction is preregistered: re-place the observed angular Petrosian radii using Eq. (30), recompute transfer/K-corrections consistently, and report the I improvement, R migration, and band spread together.

Version 7 also corrects an overrestriction in the original preregistration. Static geometry does not logically imply zero luminosity evolution because Eq. (1) is finite. Two branches are therefore preserved before data reduction: (Z) the originally registered zero-evolution test, and (E) stellar-population evolution computed independently at the static lookback time, with no evolution amplitude fitted to the Tolman residuals. If independently constrained evolution cannot reproduce the sign and magnitude of the band spread, it cannot be invoked post hoc as a rescue.

The same duality difference creates a radiance test. A localized thermal source whose spectral coordinates are shifted to $T/(1+z)$ is, before opacity, brighter by $(1+z)^2$ than a true Planck surface at that shifted temperature. Expanding-photosphere measurements of Type II supernovae at moderate redshift therefore provide a separate distance/radiance discriminator.

10 Global evolution is a real constraint, not a forbidden parameter

The observation that galaxy populations, star-formation activity, and quasar populations vary strongly with redshift is not erased by static geometry. Large surveys show pronounced evolution in the cosmic star-formation-rate density (Madau & Dickinson, 2014). Version 7 therefore does not claim a strictly stationary eternal matter distribution.

The adopted branch is: spatial geometry is static, while source populations may evolve with emission epoch. Their evolution must be described against the static lookback law Eq. (1), not fitted independently to each cosmological observable. This creates a new class of tests: stellar ages, fundamental-plane evolution, star-formation history, and quasar demographics can be compared under the static and FLRW lookback functions. At $z \sim 0.9$ the difference is modest enough that current stellar ages may not discriminate sharply; at higher redshift the histories diverge more strongly. No global population-history model is claimed here.

Quasar variability is likewise not a clean "steady-source under-dilation" kill test. Earlier work reported little or no quasar time dilation (Hawkins, 2010), while later analyses of long-baseline multi-band quasar monitoring detect a redshift-dependent variability timescale consistent with cosmological dilation (Lewis & Brewer, 2023; Brewer et al., 2025). In the present two-face model a continuously emitting source can still generate transient $\delta I(t)$ perturbations, so its variability response depends on saturation and relaxation rather than on the steady flux alone. Quasar timing therefore constrains Eq. (13); it is not yet a one-number discriminator until that response is derived.

11 Forward and non-forward energy closure

The non-forward ledger accounts explicitly for 16% rest-B beam depletion at $z = 1$, but the larger coherent energy transfer in Eq. (5) must be included in the same accounting. At $z = 1$, half the emitted photon energy has left the observed photon through the forward frequency shift. The theory therefore owes the combined ledger

$$E_{\rm emitted} = E_{\rm forward,obs} + E_{\rm nonforward} + E_{\rm bath} + E_{\rm stored}, \tag{36}$$

with the stored wake energy remaining bounded under steady illumination.

An order-of-magnitude equilibrium calculation illustrates the severity of this requirement. For $T_{\rm CMB} \simeq 2.725$ K, $u_{\rm CMB} \simeq 4.17 \times 10^{-13}$ erg cm^{-3}. Taking the optical+IR extragalactic background at roughly 50 nW m^{-2} sr^{-1} gives $u_{\rm opt} \sim 2.1 \times 10^{-14}$ erg cm^{-3}. With $K \simeq 2.27 \times 10^{-18}$ s^{-1}, the forward channel transfers energy at order

$$\dot{u}_{\rm fwd} \sim u_{\rm opt} K \sim 4.8 \times 10^{-32}\ {\rm erg\,cm^{-3}\,s^{-1}}, \tag{37}$$

about 5% of $u_{\rm CMB}$ per Hubble time. This is not a negligible throughput.

The working hypothesis is that forward and non-forward losses relax into a diffuse equilibrium bath rather than reappearing as optical halos. That immediately generates falsification gates:

(i) the non-forward differential kernel $d^2\sigma/(d\Omega\, dE')$ must not create redshift-correlated optical PSF wings or an optical/IR background excess;

(ii) the stored reactive wake must saturate and relax rather than grow without bound;

(iii) the injected non-equilibrium energy must thermalize rapidly enough to preserve the observed microwave spectrum;

(iv) the same bath dynamics must reproduce, not merely coexist with, the observed CMB anisotropy structure.

COBE/FIRAS constrains RMS deviations from a blackbody to below roughly 50 ppm of the peak, with $|y| < 15 \times 10^{-6}$ and $|\mu| < 9 \times 10^{-5}$ (Fixsen et al., 1996). If the μ limit is used only as an order-of-magnitude bound on the steady fraction of unthermalized injected energy, the above throughput requires a characteristic relaxation/thermalization time no longer than roughly

$$t_{\rm therm} \lesssim 25\ {\rm Myr}. \tag{38}$$

This is a heuristic equilibrium bound, not a substitute for a full kinetic distortion calculation. A microscopic model must compute the spectrum of the injected products and test FIRAS directly.

The CMB is not modeled here as a 3000 K last-scattering photosphere redshifted through Eq. (32); the static hypothesis treats it as the diffuse equilibrium bath itself. That avoids the naive $(1 + z)^2$ radiance mismatch for a primordial surface, but it creates a larger obligation: derive a Planck spectrum at FIRAS precision and reproduce the observed temperature/polarization acoustic peak structure, lensing, and its relation to the BAO scale. Planck's final anisotropy data are well described by six-parameter base ΛCDM(Planck Collaboration, 2020). Until an alternative bath dynamics produces comparable structure, the present model is an incomplete cosmology even if its SN transfer law survives.

12 Purpose-built photon-counting experiment

The chromatic prediction can be tested more directly than conventional classification spectroscopy permits. Once a supernova is discovered and its redshift measured, a purpose-built telescope can monitor selected observer-frame passbands corresponding to fixed rest-frame spectral windows.

A practical instrument would use a sufficiently large collecting mirror, dichroic splitting or narrow-band filters, and simultaneous APD/SPAD photon-counting channels. Red-enhanced silicon detectors cover much of the optical range; moderate-redshift monitoring of red rest-frame windows may require a near-IR counting channel. For a target at known z,

$$\lambda_{\rm filt} \simeq (1+z)\lambda_{\rm rest}. \tag{39}$$

The experiment should avoid claims about isolated Ni/Co/Fe lines in broad SN spectra. Instead it can follow integrated phase-matched windows such as the iron-group-dominated 320–340 nm region and a 620–680 nm red reference, plus an i/z/near-IR channel containing the secondary maximum.

The core observable is a simultaneous photon-count ratio

$$R_{12}(t,z) = \frac{N_1(t,z)}{N_2(t,z)}. \tag{40}$$

After throughput calibration with field standards and an internal source, the propagation law predicts

$$\frac{R_{12}(z)}{R_{12}(0)} = \exp\left\{-\tau_{\infty,B}\left[\sqrt{\frac{440}{\lambda_1}} - \sqrt{\frac{440}{\lambda_2}}\right]\left[1-(1+z)^{-1/2}\right]\right\}, \tag{41}$$

with no new opacity amplitude fitted to this experiment. The cosmological component must be approximately phase-flat at matched rest wavelengths. A sufficiently precise null therefore kills the non-forward $\lambda^{-1/2}$ channel. High cadence simultaneously improves the temporal-clock test that DES could not decide.

Before observing, the exact rest windows, filter widths, detector efficiencies, cadence, calibration-star protocol, estimator, redshift bins, and kill threshold should be preregistered. The instrument is useful precisely because it can return a meaningful null.

13 Standalone falsification ledger

Table 1: Current status of the principal claims. "Consistent" is not evidence favoring the model when FLRW predicts the same observable.

Observable/test	Current result	Status and failure condition
DES Hubble diagram	$\Delta\chi^2 = +1.1$ vs flat ΛCDM	Near degeneracy on released SN distances; SN-only result, not a global cosmological fit.
Achromatic spectral redshift	common z across spectral features	Required by coherent face; classical strongly chromatic loss mechanisms excluded. Microscopic vertex rate K still owed.
DES bulk duration	$b = 1.003 \pm 0.005 \pm 0.010$	Measures $\kappa_{\rm r} \simeq 1$; kills $b = 0$ static branches, but is shared with FLRW.
Spectral aging	aging $\propto 1/(1+z)$	Independent SN spectral clock; consistent with universal remap and FLRW.
Filter/Wien artifact	wake off: $\lvert\Delta s\rvert \le 5 \times 10^{-4}$	**Null result: proposed explanation falsified**. Chromatic erosion cannot manufacture $b = 1$.
T4 secondary maximum	$b_2 = 0.52 \pm 0.78$, $N = 120$	**Inconclusive**; contains 0 and 1. DES cadence/redshift range underpowered.

Observable/test	Current result	Status and failure condition
Raw DES color trend	-0.0963 ± 0.0140 mag/z vs intrinsic propagation $+0.019$	Opposite raw sign, but selection trend is larger; compare data-minus-published selection simulation now, then matched alternative-reference BBC.
Chromatic phase-flat ratio	$Q = 0.979/0.964/0.944$ at $z = .25/.5/1$	Open direct discriminator; amplitude fixed by Hubble fit. A precise phase-flat null kills $\lambda^{-1/2}$ channel.
Tolman distance frame	static extrapolation gives $n_I \simeq 2.30$, $n_R \simeq 1.45$	I crisis is frame fragile; R tension appears. Full per-galaxy reduction preregistered.
Tolman band spread	0.78 ± 0.21 observed vs -0.02 predicted	**Strongest current empirical threat** ($\sim 3.7\sigma$ before modern systematics/evolution calculation).
Population evolution	static lookback $K^{-1}\ln(1+z)$	V7 adopts evolving contents; must reproduce observed stellar/SFR/quasar evolution without fitting each observable independently.
Forward energy transfer	$z/(1+z)$; 50% at $z = 1$	Major open ledger. Energy destination and equilibrium kinetics must be derived.
Non-forward beam depletion	10.3% at $z = .5$, 16.0% at $z = 1$	Differential angular/spectral kernel must satisfy halo and EBL limits.
FIRAS spectrum	< 50 ppm RMS; $\lvert\mu\rvert < 9 \times 10^{-5}$	Static bath interpretation survives only if injection products thermalize rapidly; heuristic gate $t_{\rm therm} \lesssim 25$ Myr, full kinetic calculation owed.
CMB anisotropy/BAO	not computed	**Major unresolved cross-sector test**. Bath must reproduce acoustic/polarization/lensing structure and its relation to BAO.
Two-face microphysics	$\kappa_{\rm r} \simeq 1$ required by DES	Derive coherent inelastic drag, reactive transient response, saturation, relaxation, and coefficient equality from one vacuum action; otherwise model remains effective phenomenology.

14 What Version 7 establishes and what it does not

Version 7 does not establish a static universe. It establishes something narrower but nontrivial: a specific static-geometry supernova model can be written without the two immediate defects that kill classical tired light—chromatic redshift and $b = 0$ time behavior—and can be confronted with modern DES data at essentially the same SN Hubble-diagram fit quality as flat ΛCDM. The observational law is no longer an arbitrary curve: it comes with explicit timing, chromatic, angular-distance, radiance, and energy consequences.

The adversarial process also removed attractive but false support. Passband/Wien erosion does not create time dilation. T4 did not confirm the universal remap; it was underpowered. The Tolman band spread is adverse. The raw DES color trend is opposite the intrinsic chromatic prediction but presently selection dominated. The forward energy loss is larger than the non-forward loss and creates a serious CMB/thermalization debt. These are not footnotes to be edited away; they define the remaining research program.

The two-face sourcing model is the principal theoretical refinement. A per-photon coherent inelastic face can redshift steady emission, while a saturated transient reactive face can remap intervals between independent photons without requiring a globally evolving refractive index. This resolves the earlier sourcing trilemma at the level of an effective response model. It does not yet solve the microscopic problem: the same vacuum dynamics must produce the required universal K, the reactive coefficient $\kappa_{\rm r} \simeq 1$, saturation, long enough transient memory, rapid enough bath thermalization, and the correct angular/spectral redistribution. If no such local response can be derived, the model fails theoretically even if the one-dimensional SN fit remains good.

Parsimony is therefore not claimed globally. Base ΛCDMhas its own parameter and modeling structure but is already tested across the CMB, BAO, lensing, growth, and nucleosynthesis. Removing dark-energy acceleration from the SN interpretation is a parsimony gain only if the replacement propagation physics ultimately closes with comparable or lower total structure. Version 7 leaves that comparison open.

15 Conclusion

The DES-SN5YR Hubble diagram alone remains nearly indifferent between one-parameter flat ΛCDMand the one-parameter static two-channel distance law, with $\Delta\chi^2 = +1.1$. That is the empirical wedge, not the conclusion of the cosmology.

The model now has a standalone causal timing formulation. Every photon can undergo the coherent inelastic frequency shift of Eq. (2); transient changes above the saturated ambient corridor generate a separate group-delay wake, Eq. (7), shared by later independent photons. The resulting temporal exponent is $b = \kappa_{\mathrm{r}}$, and DES requires $\kappa_{\mathrm{r}} \simeq 1$. The equality of the two response rates is not yet derived microscopically and is now the central theory-side target rather than a hidden assumption.

The non-forward $E^{1/2}$ channel supplies the fitted Hubble curvature and predicts a fixed, weak chromatic erosion. Its alternative use as an explanation for time dilation was explicitly tested and falsified. The preregistered T4 species-clock measurement was honestly inconclusive. The Tolman $R - I$ spread remains the most immediate adverse optical test. The forward channel transfers 50% of photon energy by $z = 1$, making bath thermalization and CMB structure unavoidable parts of the same model rather than optional future topics.

The next useful work is therefore sharply defined: execute the full preregistered Tolman re-reduction; perform the data-minus-DES-selection color comparison and ultimately the matched alternative-reference BBC simulation; derive the two-face vertex and its energy redistribution from the vacuum action; test the bath against FIRAS and CMB anisotropies; and build the high-cadence multi-channel photon-counting experiment whose null can directly eliminate the chromatic channel.

A theory that survives only because every contradiction is reclassified is scientifically empty. Version 7 is organized to prevent that outcome: the model either closes these gates or it fails.

Data, code, and version provenance

The public DES-SN5YR light curves and v1.2 distance products are the observational basis of this work. The canonical code repository, effective-field notes, transfer-integral producers, preregistrations, T4 execution report, Tolman frame-shift analysis, forward-energy ledger, and instrument note are maintained at github.com/tracyphasespace/Static-Universe-DES-SN5YR.

Version 7 is archived at DOI 10.5281/zenodo.22031627. Prior records are intentionally preserved rather than overwritten: Draft 6 at 10.5281/zenodo.22031121 and Draft 5 at 10.5281/zenodo.22025329. The independent BBC/model-discrimination methods analysis is archived at DOI 10.5281/zenodo.22022089. These records document how adverse review, null tests, and implementation corrections changed the model.

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review, code checking, literature cross-checking, and manuscript drafting. Failed mechanisms, preregistration amendments, implementation defects, inconclusive measurements, and adverse tests are retained in the public record. The author is responsible for the hypotheses, calculations, interpretation, and final text.

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